

Introduction to Network and System Security

- Dr. Sumana Maiti

Course Name: Networks and System Security

Course Code: CS6L091

L-T-P-C: 3-0-0-3

Google Classroom code: **lkmfwzy6**

Course Teacher:

- Name: **Dr. Sumana Maiti**
- Affiliation: **Assistant Professor, CSE, IIT Bhubaneswar**
- Contact Information: sumana@iitbbs.ac.in
- Research Background:
 - Cryptography,
 - Post-Quantum Cryptography,
 - AI in Cyber Security
 - Game Theory

Teaching Assistant

- **Bibek Giri (PhD)**
- **Bidisha Chakraborty (PhD)**
- **Rahul Dewangan (M.Tech CS)**
- **Konisa Sai Sriyuktha (M.Tech CS)**
- **Arindam Das (M.Tech AI)**

Key Topics to be Covered

- ❑ Basics of Computer Network: [OSI Layer](#), Popular protocols of [Data link layer](#), [Network layer](#), [Transport layer](#), and [Application layer](#), Client Server and peer to peer architectures.
- ❑ Computer Security Concepts , OSI Security Architecture, Security Attacks, Security Services, [CIA Principles](#), Security Mechanisms.
- ❑ [Network Attacks](#): Data link layer attacks, Network layer attacks, Transport layer attacks, Application layer attacks
- ❑ Firewall and [Access Control](#): Different types of access control mechanism, Working principle of [Firewall](#).
- ❑ Introduction to [Cryptography: Symmetric key cryptography](#)- Block Cipher, Stream Cipher, [DES](#), 3DES, [AES](#), Security analysis of symmetric key cryptography, [Diffie-Hellman algorithm](#)
- ❑ Public key cryptography: [RSA](#), [Elgamal](#), [Elliptic curve cryptography](#), [Digital Signature](#), [PKI](#), Analysis of public key cryptography.
- ❑ IP Security and [Intrusion Detection System](#)

Course Learning Outcome

- Describe fundamental security concepts such as confidentiality, integrity, availability, authentication, and non-repudiation in network and system environments.
- Apply basic cryptographic techniques (symmetric/asymmetric encryption, hashing, digital signatures) and security mechanisms to protect data and communications.
- Identify, analyze, and describe common network and system threats, attacks (e.g., malware, DoS, MITM, phishing), and their underlying vulnerabilities.
- Design and implement security controls such as access control models, authentication schemes, monitoring, and simple incident-response procedures in networked systems.

Book Name

1. William Stallings, “Network Security Essentials Applications and Standards”
2. Kaufman, Perlman and Speciner, “Network Security: Private Communication in a Public World”
3. B. Forouzan, “Introduction to Cryptography and Network Security”
4. B. Forouzan, “Data Communication and Networking”

Evaluation Strategy and Attendance Policy

Evaluation Strategy

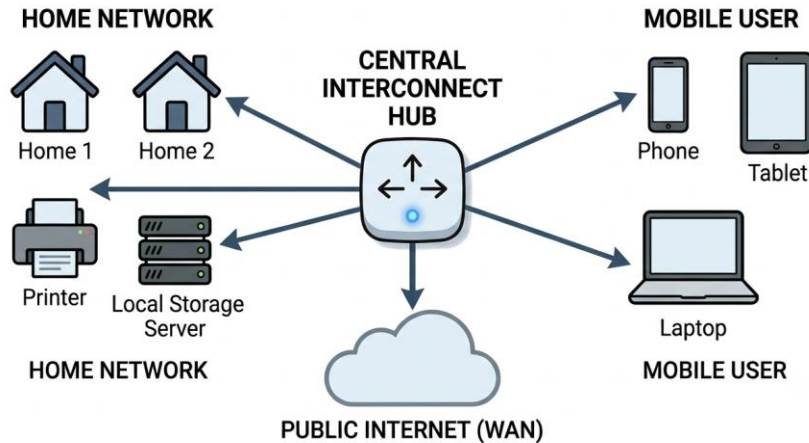
- Mid Semester: 25 to 30 Marks
- End Semester: 45 to 50 Marks
- Class Participation: 5 to 10 Marks
- Active Collaborative Learning: 10 Marks

Attendance

- 5 marks are set aside for attendance. A student is given attendance marks varying linearly from -5 for 0% attendance to +5 for 100% attendance. As per this formula, students get credit for attendance > 50% and debit if it is less.
- Faculty advisor will be informed if consecutive absent for two days

What is Computer Network?

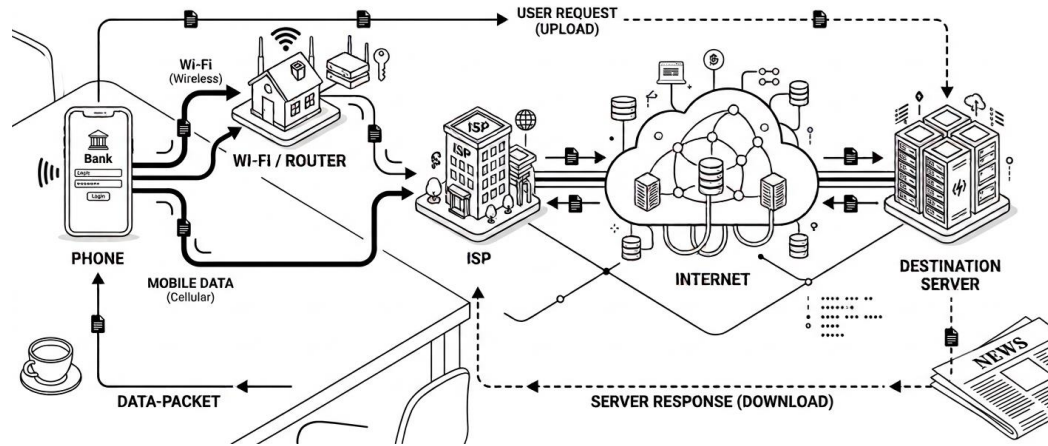
“A network is a group of two or more connected devices that communicate and share data, resources, or services.”



Why Network and System Security?

Protecting computers, servers, and networks from unauthorized access, attacks, and damage.

- How the attack works on the network level
- How to design systems so that it can't happen



Need to ensure

1. Who can read my message
2. Can someone change it without my knowledge
3. Can someone make the server stop working?

Free Wifi, Log into bank account, Someone in the same network silently monitoring

Everytime you send a message or pay online, there are some invisible protocols for routing, error checking, and establishing trust.
So, in this course we will learn

- *How does a message find its way across the world?*
- *How do devices agree on addresses?*
- *How do we know that we are talking to the right server?*
- *How do the attacker intercepts, redirect, and disrupt the communication?*
- *How do the defenders detect, prevent, and respond to the attacks?*